

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2018
 Programme: BE Full Marks: 100
 Course: Microprocessor and Assembly Language Programming Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the evolution of Intel Series Microprocessors starting from 16-bit architecture to 64-bit architectures. 7
 b) Draw block diagram of a Microprocessor and explain it in detail. 8
2. a) Write the function, addressing modes, size and name of machine cycles for the following instructions: i) LDAX D ii) ADI 32H iii) LXI B, 2075H 6
 b) Define Instruction cycle, Machine cycle and T-states. 3
 c) Draw the timing diagram for the 8085 instruction MVI A, 32H. 6
3. a) What do you mean by Segmentation Offset Scheme in 8086 Microprocessor. Explain with suitable example. 5
 b) Write an 8085 ALP to subtract two 16-bit numbers and store the result in memory locations starting from 2075H. 5
 c) Explain various assembler directives in brief. 5
4. a) Write an assembly language program to find the sum of two number which is input by the user through the key board and display the result in screen. 7
 b) Write a procedure program for 8086 for newline and use it to display three different strings in different lines. 8
5. a) What are various sources of interrupts? Explain interrupt vector table of 8086 microprocessor. 8
 b) What do you mean by address decoding? Design an address decoding circuit to interface 4K×8 RAM, 8K×8 ROM and 16K×8 RAM with starting address 0000h. 7
6. a) You are given a microprocessor system with clock frequency 10MHz. Write a program for 8254 PIT to generate a square wave of frequency 2KHz. 5

- b) How cascading is done to handle more than 8 interrupts using 8259 PICs? Explain. 5
- c) Write 8085 program for 8255 PPI to take input from input device connected to Port B and display the value of input on the output device connected to Port A of 8255 PPI. 5
7. Write short notes on: (Any two) 2×5
 - a) Direct Memory Access Controller
 - b) Different types of Assemblers
 - c) 8085 flag register